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Sub.: THERMOPAC LUBRICANT OIL BLENDING PLANT OFFER 100TPD – 15,000 KLPA

Kind Attn: Mr. Khalid Alhajri

Dear Sir

We refer to our discussion on. 11th Feb. regarding the Proposed Lube oil Blending Plant.

We are pleased to submit our final offer for design, manufacture, supply and commissioning of lube oil blending plant to be installed and commissioned at your site.

The plant is designed with the latest technology, with a view to maximize blending flexibility by means of environment friendly batch process with minimum risk of product contamination and rework.

The process and its advantages along with technical details are described in the following pages.



For Thermopac Process engineering LLP

Manager Marketing

1. Project Overview

Lubricant Blending and Packaging Plant

1.1 Introduction

THERMOPAC proposes to design, engineer, supply, install, and commission a **complete Lubricant Blending and Packaging Plant** capable of producing a wide range of automotive and industrial lubricants in compliance with international standards such as **API, SAE, ASTM, and OEM specifications**.

The facility will integrate **raw material storage, automated blending systems, finished product storage, packaging lines, and plant-wide automation**, ensuring reliable production, high product quality, and optimized operating costs.

The proposed plant will be designed using modern blending technologies and automated control systems to achieve **high blending accuracy, rapid batch turnaround, minimal contamination, and efficient resource utilization**.

The plant can process **base oils from conventional refineries or re-refined base oils**, enabling investors to produce finished lubricants from recycled feedstock and significantly increase value addition.

1.2 Project Objectives

The primary objective of the project is to establish a **modern lubricant manufacturing facility** capable of producing a diversified product portfolio while maintaining high operational efficiency and product quality.

Key objectives include:

- Production of **high-quality lubricants meeting international specifications**
- Flexible manufacturing to handle **multiple formulations and batch sizes**
- Automation to ensure **accurate dosing and repeatable blending**
- Efficient material handling to minimize contamination and product losses
- Optimized plant layout to reduce operating costs and manpower requirements
- Capability to produce both **bulk and packaged lubricants**

1.3 Plant Capacity

The proposed plant is designed with a nominal production capacity of:

100 Metric Tons per Day (MTD)

or approximately

30,000 Metric Tons per Year (TPA) (based on 300 operating days).

The plant configuration will allow flexible operation with multiple blending batches per day depending on product types and packaging requirements.

1.4 Product Portfolio

The plant will be capable of manufacturing a wide range of lubricants including:

Automotive Lubricants

- Engine oils (mineral and synthetic blends)
- Transmission oils
- Gear oils
- Motorcycle oils
- ATF fluids

Industrial Lubricants

- Hydraulic oils
- Turbine oils
- Compressor oils
- Circulating oils
- Heat transfer oils

Specialty Lubricants

- Marine lubricants
- Grease base oils
- Specialty industrial lubricants

Products will be available in both **bulk and packaged formats**, including:

- 200 L drums
- 20 L pails
- 4 L containers
- 1 L bottles
- bulk tanker loading

1.5 Plant Configuration

The lubricant manufacturing facility will consist of the following major sections:

1. Raw Material Storage Section

- Base oil storage tanks
- Additive storage systems
- Drum handling and additive dosing systems

2. Blending Section

- Automated blending kettles
 - Agitation and heating systems
 - Batch control systems
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3. Intermediate Storage (Holdup Tanks)

- Holdup / intermediate storage tanks for blended products
- Buffer tanks between blending and packaging operations
- Transfer pumps and pipelines

Holdup tanks allow the plant to **decouple blending operations from packaging operations**, enabling continuous production and improving plant efficiency.

4. Finished Product Storage

- Dedicated finished product storage tanks
- Bulk product transfer pumps and pipelines

5. Packaging Section

- Drum filling lines
- Small pack filling lines
- Labeling and packaging systems

6. Utilities and Support Systems

- Thermic fluid heating system
- Compressed air system
- Electrical distribution system
- Fire protection system

7. Automation and Control

- PLC / SCADA based plant control
- Batch recipe management
- Production monitoring and reporting

1.6 Process Philosophy

The plant will utilize a **controlled batch blending process**, allowing flexible manufacturing of multiple lubricant formulations with high blending accuracy and repeatability.

The process will involve:

1. Transfer of base oils from storage tanks to blending vessels
2. Controlled heating to blending temperature
3. Precise dosing of additive packages
4. Agitation and homogenization of the mixture
5. Quality verification and filtration
6. Transfer to finished product storage tanks
7. Packaging or bulk dispatch

Automation systems will ensure **accurate formulation control, minimal operator intervention, and full traceability of production batches.**

1.7 Key Design Considerations

The plant design will be developed considering the following criteria:

- flexibility for multiple lubricant grades
- optimized batch cycle time
- high blending accuracy
- contamination control
- efficient material handling
- energy efficiency
- compliance with international safety standards

The facility layout and process configuration will be optimized to ensure **maximum productivity, operational reliability, and future expandability.**

1.8 Integration with Re-Refining Plants

For clients operating **used oil re-refining facilities**, the blending plant can be integrated directly with the base oil production unit, allowing the production of finished lubricants from recovered base oils.

This integrated configuration provides:

- higher product value
- improved profitability
- reduced dependence on external base oil suppliers
- complete circular oil recycling solution

Process Design Philosophy

Lubricant Blending and Packaging Plant

2.1 Design Approach

The lubricant blending facility will be designed as an **integrated production system** combining raw material storage, blending operations, product transfer, packaging, and plant-wide automation.

The design philosophy is based on the following principles:

- High blending accuracy and repeatability
- Flexible production for multiple lubricant grades
- Efficient material handling with minimal contamination
- Reduced batch cycle time and optimized throughput
- High level of automation and process monitoring
- Compliance with international engineering and safety standards

The process configuration will ensure that the plant can operate reliably under varying production requirements while maintaining consistent product quality.

2.2 Raw Material Handling Philosophy

Lubricant manufacturing primarily involves blending **base oils and additive packages** in precise proportions.

The plant will be designed to handle the following raw materials:

Base Oils

- Group I base oils (SN70, SN150, SN300, SN500)
- Group II / Group III base oils
- Re-refined base oils (if integrated with used oil re-refining plants)

Additives

- Viscosity Index Improvers (VII)
- Detergent and dispersant packages
- Anti-wear additives
- Anti-oxidants
- Pour point depressants
- Corrosion inhibitors

Raw materials will be received in bulk tankers, IBC containers, or drums and transferred to dedicated storage tanks or dosing systems.

The plant design will ensure:

- controlled storage conditions
- efficient transfer systems
- accurate dosing into blending vessels

2.3 Blending Philosophy

The plant will utilize **automated batch blending technology**, which is the most widely used approach for lubricant production due to its flexibility and formulation accuracy.

Batch blending is particularly suitable for plants producing a wide range of lubricant grades with different formulations.

The blending process includes:

1. Charging base oil into blending vessels
2. Heating the base oil to the required blending temperature
3. Controlled dosing of additive packages
4. Mechanical agitation for homogenization
5. Quality verification through sampling and testing
6. Transfer of finished product to storage tanks

Automated blending systems ensure:

- accurate dosing of base oils and additives
- repeatable product formulations
- minimized operator intervention
- traceability of production batches

2.4 Blending Technology Selection

The final blending configuration will be selected based on the client's production portfolio and capacity requirements.

Possible blending technologies include:

Automatic Batch Blending

- Suitable for complex formulations
- High production flexibility
- Ideal for multiple product grades

In-Line Blending

- Suitable for high throughput operations
- Continuous production capability
- Minimal intermediate storage

Hybrid Blending Systems

- Combination of batch and in-line blending
- Allows both small and large batch production
- Provides maximum operational flexibility

The plant design will prioritize the blending method that provides the best balance between **flexibility, productivity, and capital investment**.

2.5 Process Temperature Control

Temperature control is critical in lubricant blending operations to ensure proper dissolution and dispersion of additive packages.

Heating systems will be provided for:

- base oil heating
- additive dissolution
- viscosity control during blending

The plant will utilize a **thermic fluid heating system** capable of delivering stable process temperatures up to approximately **250–300°C**, ensuring reliable heating performance with safe operation.

Temperature monitoring and control will be integrated with the plant automation system.

2.6 Contamination Control

Preventing cross-contamination between lubricant grades is essential to maintain product quality.

The plant design will incorporate several contamination control measures:

- dedicated pipelines for critical products
- flushing procedures between batches

- filtration systems prior to storage and filling
- controlled transfer sequences

Advanced plants may also incorporate **pipeline pigging systems** to eliminate product losses and reduce contamination during transfers.

2.7 Product Transfer Philosophy

Finished lubricants will be transferred from blending vessels to product storage tanks using dedicated pumps and pipelines.

The transfer system will be designed to ensure:

- reliable product flow
- minimal pressure losses
- controlled transfer rates
- prevention of product degradation

Transfer pipelines will be designed to facilitate efficient cleaning and maintenance.

2.8 Filtration and Product Quality Assurance

Prior to storage or packaging, the blended lubricant will pass through filtration systems to remove any impurities or particulate matter.

Typical filtration levels:

- 5–20 micron filtration for finished lubricants

Quality control procedures will include:

- viscosity measurement
- flash point testing
- density testing
- additive concentration verification

These tests ensure that the final product meets the required international standards.

2.9 Automation Philosophy

The plant will be equipped with a **PLC / SCADA-based automation system** to ensure efficient plant operation.

Automation functions will include:

- recipe management
- batch sequencing
- dosing control
- temperature control
- production monitoring
- alarm management

The automation system will provide:

- full traceability of production batches
- real-time plant monitoring
- improved operational safety
- reduced manpower requirements

2.10 Process Safety Considerations

The plant design will incorporate safety measures to ensure reliable and safe operation.

Key safety features include:

- temperature and pressure monitoring
- emergency shutdown systems
- fire protection systems
- safe handling of flammable liquids
- electrical grounding and static protection

All systems will be designed in accordance with internationally accepted engineering and safety standards.

2.11 Expandability and Future Integration

The plant layout and process design will allow for future expansion of capacity or integration of additional process units.

Potential expansion options include:

- additional blending vessels
- expanded storage tank capacity
- additional packaging lines
- integration with base oil re-refining units

This approach ensures long-term scalability of the facility while minimizing future capital expenditure.

Design Codes and Standards

The design, engineering, fabrication, installation, and commissioning of the lubricant blending plant will comply with internationally recognized engineering codes and standards to ensure safety, reliability, and product quality.

The following standards will be applied where applicable:

Standard	Description
ASTM	American Society for Testing and Materials – product testing and lubricant quality standards
ASME	American Society of Mechanical Engineers – design and fabrication of pressure equipment
API	American Petroleum Institute – petroleum equipment and storage standards
TEMA	Tubular Exchanger Manufacturers Association – heat exchanger design standards
BIS	Bureau of Indian Standards – applicable national engineering standards
Local Codes	Applicable statutory and safety regulations of the country where the plant is installed

Where multiple standards apply, the most stringent requirements will be adopted to ensure safe and reliable plant operation.

4. Advantages of the Proposed Process

The proposed lubricant blending plant incorporates modern engineering practices and advanced automation systems to ensure efficient and reliable operation.

The key advantages of the process include:

Environmentally Friendly Operation

The process is designed to minimize waste generation and reduce environmental impact through efficient material handling and controlled blending operations.

High Process Efficiency

The plant utilizes modern blending technology and optimized process design to achieve high productivity and efficient use of raw materials.

Minimal Contamination

Dedicated transfer systems, filtration units, and controlled blending sequences ensure minimal cross-contamination between lubricant grades.

Advanced Automation

The plant is equipped with a **PLC-based automation system with SCADA monitoring**, enabling precise process control, reduced manpower requirements, and improved operational reliability.

Modular and Skid-Mounted Design

Major process units can be supplied as **skid-mounted modules**, allowing faster installation, reduced construction time, and simplified plant expansion.

Expandable Capacity

The modular design allows easy expansion of production capacity by adding additional blending vessels, storage tanks, or packaging lines.

Flexible Product Handling

The plant is capable of producing a wide range of lubricant products including automotive oils, industrial oils, specialty lubricants, and synthetic blends.

3 Plant Capacity and Product Slate

This section will include:

- production capacity
- batch sizes
- product mix
- packaging formats
- annual production calculations

Plant Capacity & Product Slate

Lubricant Blending and Packaging Plant

3.1 Plant Production Capacity

The proposed lubricant manufacturing facility is designed with a **nominal production capacity of 100 Metric Tons per Day (MTD)**.

Assuming typical operating conditions:

Parameter	Value
Daily production capacity	100 MT/day
Operating days per year	300 days
Annual production capacity	30,000 MT/year

The plant will operate on a **batch blending basis**, with multiple blending cycles per day depending on the product formulation and packaging schedule.

3.2 Blending Batch Size

The blending section will consist of multiple blending vessels designed to provide operational flexibility.

Typical design configuration:

Equipment	Quantity	Capacity
Blending vessels	4–6 units	10 KL each
Batch production per vessel	2–3 batches/day	20–30 KL/day
Total blending capacity	–	~100 MT/day

This configuration allows:

- simultaneous production of different lubricant grades
- flexibility in batch sizes
- efficient production scheduling

3.3 Product Portfolio

The plant will be capable of producing a wide range of **automotive and industrial lubricants**.

Automotive Lubricants

Product	Typical Specification
Passenger Car Motor Oil	API SN / SP
Diesel Engine Oil	API CI-4 / CK-4
Motorcycle Oil	JASO MA
Automatic Transmission Fluid	ATF
Gear Oil	API GL-4 / GL-5

Industrial Lubricants

Product	Typical Standard
Hydraulic Oil	DIN 51524
Compressor Oil	ISO 6743
Turbine Oil	ISO 8068
Circulating Oil	ISO 11158
Heat Transfer Oil	ASTM standards

Specialty Lubricants

Product	Application
Marine lubricants	Marine engines
Transformer oils	Electrical equipment
Metalworking fluids	Industrial machining
Synthetic lubricants	High temperature applications

3.4 Base Oil Requirements

Lubricant blending typically consists of **85–90% base oil and 10–15% additive packages**.

Typical base oils used:

Base Oil Type	Typical Use
SN150	Light lubricants
SN300	Medium viscosity oils
SN500	Heavy duty lubricants
Bright Stock	High viscosity lubricants
Group II / III	Premium lubricants

The plant can also process **re-refined base oils** produced from used oil recycling plants.

3.5 Additive Packages

Additives enhance lubricant performance and typically represent **10–15% of the formulation**.

Common additives include:

Additive	Function
Viscosity Index Improver	Maintain viscosity stability
Detergents	Prevent deposits
Dispersants	Keep contaminants suspended
Anti-wear additives	Protect metal surfaces
Anti-oxidants	Prevent oil degradation
Pour point depressants	Improve low temperature flow

Additives will be stored in **dedicated tanks, IBC containers, or drums** and dosed into blending vessels using controlled dosing systems.

3.6 Packaging Formats

Finished lubricants will be supplied in both **bulk and packaged formats**.

Packaging Type	Typical Application
Bulk tanker loading	Industrial customers
200 L drums	Industrial supply
20 L pails	Commercial supply
5 L containers	Automotive retail
1 L bottles	Retail market

Packaging lines will be designed to support **simultaneous filling of multiple container sizes**.

3.7 Packaging Capacity

Typical packaging line capacities:

Container Size	Filling Capacity
1 L bottles	40–60 bottles/min
4–5 L containers	20–25 containers/min
20 L pails	300–400 units/hour
200 L drums	25–30 drums/hour

Multiple packaging lines can operate simultaneously depending on production demand.

3.8 Storage Philosophy

The plant will include storage facilities for:

Raw Materials

- Base oil tanks
- Additive tanks
- Drum storage

Finished Products

- Bulk product storage tanks
- Intermediate holding tanks
- packaged product warehouse

Typical tank capacities will be designed to provide **7–15 days of production storage** to ensure uninterrupted operation.

3.9 Production Flexibility

The plant will be designed to support:

- rapid product changeovers
- multiple lubricant grades
- varying batch sizes
- flexible packaging formats

This flexibility allows the plant to respond quickly to market demand while maintaining high production efficiency.

3.10 Integration with Re-Refining Plants

If the blending plant is integrated with a **used oil re-refining facility**, the produced base oil can be directly used for lubricant manufacturing.

Benefits include:

- higher value addition
- reduced base oil procurement costs
- improved supply chain control
- sustainable circular oil recycling

Offered Plant Configurations

Lubricant Blending Plant – Standard Capacities

THERMOPAC offers a range of **modular lubricant blending plants** designed for different production capacities. These plants are supplied as **fully automated PLC–SCADA controlled turnkey projects**, including engineering, equipment supply, installation, and commissioning.

The offered plant configurations are summarized below.

Product Code	Description	Nominal Capacity
LOB-APS-CTP-50	Lube Oil Blending Plant – Automatic PLC/SCADA Controlled Turnkey Project	50 TPD
LOB-APS-CTP-100	Lube Oil Blending Plant – Automatic PLC/SCADA Controlled Turnkey Project	100 TPD
LOB-APS-CTP-150	Lube Oil Blending Plant – Automatic PLC/SCADA Controlled Turnkey Project	150 TPD
LOB-APS-CTP-200	Lube Oil Blending Plant – Automatic PLC/SCADA Controlled Turnkey Project	200 TPD

4.1 Modular Design Philosophy

All plant configurations are based on a **modular engineering concept**, allowing capacity expansion with minimal disruption to plant operations.

Key features of the modular design include:

- scalable blending capacity
- expandable tank farm and storage systems
- modular automation architecture
- adaptable packaging line capacity
- simplified installation and commissioning

This approach allows clients to **start with smaller capacity plants and expand production as market demand grows**.

4.2 Typical Plant Capacity Comparison

Parameter	50 TPD	100 TPD	150 TPD	200 TPD
Annual Production (300 days)	15,000 MT	30,000 MT	45,000 MT	60,000 MT
Typical Blending Vessels	2–3	4–6	6–8	8–10
Base Oil Storage	500–800 KL	800–1200 KL	1200–1800 KL	1800–2500 KL
Installed Power	~150 kW	~300 kW	~450 kW	~600 kW
Land Requirement	~2,500 m ²	~4,000 m ²	~6,000 m ²	~8,000 m ²

4.3 Automation and Control

All plant configurations include:

- **PLC based process control**
- **SCADA monitoring system**
- recipe management
- batch production control
- real-time plant monitoring
- production reporting

The automation system ensures:

- precise additive dosing
- repeatable product quality
- reduced manpower requirement
- full traceability of production batches

4.4 Turnkey Scope of Supply

The turnkey project scope includes:

Engineering

- Process design
- PFD & P&ID preparation
- plant layout and tank farm design
- utility system design

Equipment Supply

- blending vessels
- pumps and pipelines
- heating system
- filtration system
- storage tanks
- filling and packaging lines

Automation

- PLC control panels
- SCADA system
- instrumentation and sensors

Installation and Commissioning

- mechanical erection
- piping installation
- electrical installation
- plant commissioning and operator training

✓ Next we should create a **very powerful section used by EPC companies:**

5 Key Features of the Offered Plant

This section is **very important for sales** because it will highlight advantages like:

- advanced blending technology
- low contamination design
- high automation
- energy efficiency
- integration with re-refining plants

Equipment Description

Lubricant Blending and Packaging Plant

The proposed lubricant blending facility will include a complete set of process equipment designed for reliable, efficient, and flexible operation. The equipment will be selected and engineered to ensure high blending accuracy, operational safety, and ease of maintenance.

The major equipment included in the plant is described below.

5.1 Base Oil Storage Tanks

Base oil storage tanks are provided for receiving and storing different grades of base oils prior to blending operations.

Typical base oils handled include:

- SN150
- SN300
- SN500
- Bright Stock
- Group II / Group III base oils
- Re-refined base oils (if integrated with recycling plants)

Design Features

- Vertical cylindrical storage tanks
- Carbon steel construction
- Bottom drain connections
- Level measurement instrumentation
- Heating coils (if required)
- Tank venting and safety fittings

Typical Capacity

Tank Type	Typical Capacity
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Base Oil Storage Tanks	100 – 500 KL each
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5.2 Additive Storage and Handling System

Additives are stored in dedicated tanks, drums, or IBC containers and transferred to the blending vessels using controlled dosing systems.

Typical Additives

- Detergents and dispersants
- Anti-wear additives
- Anti-oxidants
- Viscosity index improvers
- Pour point depressants
- Corrosion inhibitors

System Components

- Additive storage tanks
 - Drum decanting systems
 - Additive transfer pumps
 - Metering and dosing systems
-

5.3 Blending Vessels

Blending vessels are the core equipment used for mixing base oils and additives to produce finished lubricants.

Design Features

- Cylindrical steel vessels with dished bottom
- Top-mounted agitator
- Thermic fluid heating jacket
- Temperature monitoring
- Load cell weighing system
- Sampling valves
- Bottom discharge system

Typical Configuration

Equipment	Quantity	Capacity
Blending vessels	4 – 6	10 KL each

Each vessel is designed to produce **multiple batches per day** depending on the product formulation.

5.4 Agitation System

Blending vessels are equipped with mechanical agitators designed to provide efficient mixing and homogenization.

Features

- Turbine type agitator blades
- Gear-driven electric motor
- Variable speed operation (optional)
- Heavy-duty shaft and bearing assembly

The agitation system ensures rapid mixing and uniform distribution of additive packages throughout the base oil.

5.5 Thermic Fluid Heating System

Heating is required to maintain proper blending temperature and ensure effective dissolution of additives.

The plant includes a **thermic fluid heater system** to supply heat to the blending vessels.

Typical Specifications

Parameter	Typical Value
Heating capacity	~1,000,000 kcal/hr
Operating temperature	up to 300°C
Fuel type	Diesel / Furnace Oil / Gas
Thermal efficiency	~85–90%

System Components

- Thermic fluid heater
 - circulation pumps
 - expansion tank
 - thermic fluid piping network
 - temperature control system
-

5.6 Transfer Pumping System

A network of pumps is provided to transfer base oils, additives, and finished products between different process sections.

Pump Types

- Base oil transfer pumps
- Additive dosing pumps
- product recirculation pumps
- finished product transfer pumps

Typical Capacity

Pump Service	Flow Rate
Base oil charging pump	15–20 m ³ /hr
Recirculation pump	8–10 m ³ /hr
Finished product transfer	10–20 m ³ /hr

All pumps are designed for reliable operation with lubricants of varying viscosity.

5.7 Filtration System

Filtration units are installed to remove particulate contaminants before storage and packaging.

Features

- Basket filters
- cartridge filters
- stainless steel filter elements

Typical Filtration Range

Filtration Level	Application
5 – 20 micron	Finished lubricants

This ensures the final product meets quality requirements.

5.8 Finished Product Storage Tanks

Finished lubricants are stored in dedicated tanks prior to packaging or bulk dispatch.

Design Features

- Carbon steel construction
- level measurement instrumentation
- product transfer connections
- tank venting and safety devices

Typical Capacity

Tank Type	Capacity
Finished Product Tanks	50 – 200 KL

5.9 Packaging and Filling System

The packaging section is designed to fill finished lubricants into various container sizes.

Packaging Formats

- 1 L bottles
- 4 L containers
- 20 L pails
- 200 L drums

Equipment Included

- small pack filling machines
- drum filling machines
- automatic capping machines
- induction sealing machines
- labeling machines
- carton sealing machines

These systems ensure efficient and accurate packaging operations.

5.10 Material Handling Equipment

Material handling equipment is provided to facilitate movement of raw materials and finished products.

Typical equipment includes:

- hydraulic drum lifters
 - pallet trucks
 - forklifts (optional)
 - drum trolleys
-

5.11 Air Compressor System

Compressed air is required for operation of pneumatic valves, packaging equipment, and plant utilities.

Typical Specification

Parameter	Value
Compressor capacity	~30 CFM
Working pressure	6 – 12 bar
Receiver tank	~500 L

5.12 Instrumentation

Instrumentation ensures reliable monitoring and control of process parameters.

Typical instruments include:

- level transmitters
- temperature transmitters
- pressure gauges
- flow meters
- load cells
- control valves

All instrumentation is integrated with the plant automation system.

Next Section

The next section should be:

6 Automation & Control Philosophy

This is **one of the most important sections** because modern lubricant plants are **highly automated**, and it is where companies like ABB differentiate themselves.

In that section we will define:

- PLC architecture
- SCADA system
- batch recipe management
- production reporting
- remote monitoring

Automation Philosophy

Lubricant Blending and Packaging Plant

6.1 Automation Concept

The lubricant blending plant will be equipped with a **fully integrated PLC–SCADA based automation system** designed to ensure precise process control, operational reliability, and full traceability of production batches.

The automation system will control and monitor all major plant operations including:

- raw material receipt and storage
- base oil transfer and heating
- additive dosing
- blending operations
- product transfers
- packaging operations
- utilities monitoring

The system architecture will provide **centralized monitoring and decentralized control**, ensuring efficient plant operation with minimal operator intervention.

6.2 Automation Architecture

The plant automation system will consist of three primary levels:

Level 1 – Field Level

This level includes all field instrumentation and devices such as:

- level transmitters
- temperature transmitters
- pressure transmitters
- flow meters
- load cells for blending vessels
- control valves
- motor control interfaces

These devices provide real-time process data and execute control commands.

Level 2 – PLC Control Level

The programmable logic controller (PLC) will perform real-time control of plant operations including:

- blending sequence control
- additive dosing control
- heating temperature regulation
- pump operation control
- interlock and safety systems

The PLC ensures that all operations are performed in the correct sequence and within defined operating parameters.

Level 3 – SCADA / Supervisory Level

The Supervisory Control and Data Acquisition (SCADA) system will provide the human-machine interface (HMI) for plant operators.

Functions include:

- real-time process visualization
- batch monitoring
- alarm management
- historical data logging
- production reporting

Operators will be able to monitor and control the entire plant from a centralized control room.

6.3 Recipe Management System

The automation system will include a **recipe management module** for lubricant formulations.

This system allows:

- storage of multiple lubricant formulations
- automatic dosing of base oils and additives
- precise control of blending sequences
- repeatable production of lubricant grades

Recipe-based control ensures **consistent product quality and elimination of manual formulation errors**.

6.4 Batch Production Control

The plant will operate on a **batch production system** where each blending cycle is controlled automatically by the PLC.

Batch operations will include:

1. Charging of base oil into blending vessel
2. Controlled heating to blending temperature
3. Sequential dosing of additives
4. Agitation and homogenization
5. sampling and quality verification
6. transfer of finished product to storage tanks

The automation system will manage the entire sequence automatically once the recipe is selected.

6.5 Alarm and Safety Interlocks

The automation system will include a comprehensive alarm and interlock system to ensure safe plant operation.

Typical safety functions include:

- high temperature alarms
- high level tank protection
- pump interlocks
- emergency shutdown system
- motor overload protection

All alarms will be displayed on the SCADA system and recorded in the event log for operator review.

6.6 Production Monitoring and Reporting

The automation system will continuously collect and store production data.

Typical reports include:

- batch production reports
- raw material consumption
- additive usage
- production volume by product grade
- equipment operating status

These reports provide valuable information for plant management and operational optimization.

6.7 Remote Monitoring Capability

The automation system can be configured with **remote monitoring capabilities**, allowing plant performance to be monitored from remote locations.

This feature enables:

- technical support from equipment suppliers
- remote troubleshooting
- performance analysis
- operational optimization

Secure communication protocols will be used to ensure data integrity and system security.

6.8 Integration with Business Systems

The automation system can be integrated with external systems such as:

- ERP systems
- laboratory information systems
- inventory management systems
- warehouse management systems

Integration allows automatic exchange of production data and improves overall plant operational efficiency.

6.9 Advantages of the Automation System

The automation philosophy provides the following operational benefits:

- precise additive dosing and blending accuracy
- consistent product quality
- reduced manpower requirements
- improved operational safety
- full traceability of production batches
- enhanced plant productivity

This automation approach ensures that the lubricant blending facility operates as a **modern, efficient, and highly controlled manufacturing system**.

Next Section

The next EPC section should be:

7 Utility Systems

This will include:

- Thermic fluid heating system
- Compressed air system
- Electrical distribution
- Water systems
- Fire protection

EPC Scope of Work

Engineering, Procurement and Construction

THERMOPAC will execute the lubricant blending plant as a **complete turnkey project**, covering engineering, equipment supply, installation supervision, and commissioning services.

The scope of work is structured into the following major phases.

7.1 Engineering Scope

THERMOPAC will perform all required engineering activities necessary for the successful implementation of the lubricant blending plant.

Process Engineering

The process engineering scope includes:

- Process design basis
 - Process Flow Diagram (PFD)
 - Piping and Instrumentation Diagrams (P&ID)
 - equipment sizing and selection
 - material balance calculations
 - utility consumption calculations
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Mechanical Engineering

Mechanical engineering activities include:

- design of blending vessels
 - storage tank design
 - pump selection
 - piping layout and routing
 - equipment fabrication drawings
 - equipment specifications
-

Electrical Engineering

Electrical engineering includes:

- electrical load calculations
- motor control center (MCC) design
- power distribution system
- cable routing diagrams

- grounding and earthing systems
- lighting systems

Instrumentation and Control Engineering

Instrumentation engineering includes:

- instrumentation specification
- control system architecture
- PLC–SCADA system design
- instrument loop diagrams
- control panel design

Plant Layout Engineering

Engineering will include preparation of:

- general arrangement drawings
- equipment layout
- piping layout
- utility layout
- packaging area layout

7.2 Procurement Scope

THERMOPAC will supply all major equipment and materials required for the plant.

Major Process Equipment

The supply includes:

- blending vessels
- base oil storage tanks
- additive storage systems
- thermic fluid heating system
- transfer pumps
- filtration systems
- finished product storage tanks

Packaging Equipment

The packaging section will include supply of:

- small pack filling machines
- drum filling machines
- capping machines
- induction sealing machines
- labeling machines

- carton sealing machines

Utilities Equipment

Utility systems will include:

- thermic fluid heater
- circulation pumps
- air compressor
- electrical panels
- instrumentation

Automation System

The automation system will include:

- PLC control panels
- SCADA system
- HMI operator stations
- field instrumentation
- control valves and sensors

7.3 Construction and Installation Scope

THERMOPAC will provide technical supervision and support during plant installation.

Installation scope includes:

- equipment erection
- piping installation
- electrical installation
- instrumentation installation
- insulation works
- plant integration

All installation work will be carried out in accordance with approved engineering drawings and safety standards.

7.4 Pre-Commissioning

Before plant startup, the following activities will be performed:

- mechanical completion verification
- equipment alignment and inspection
- piping pressure testing
- electrical system checks
- instrumentation calibration
- automation system testing

These activities ensure that all equipment is ready for safe plant startup.

7.5 Commissioning and Startup

THERMOPAC commissioning engineers will supervise plant startup and ensure proper system operation.

Commissioning activities include:

- initial system energization
- utility system startup
- blending system testing
- automation system validation
- trial production runs

The plant will be operated under supervision until stable operation is achieved.

7.6 Performance Testing

After commissioning, performance testing will be conducted to verify that the plant meets the required operational parameters.

Testing will verify:

- blending capacity
- automation performance
- equipment functionality
- product quality

Successful completion of performance testing will confirm plant acceptance.

7.7 Training and Technical Support

THERMOPAC will provide comprehensive training for the client's operating personnel.

Training will include:

- plant operation procedures
- equipment maintenance
- automation system operation
- safety procedures

Technical support will also be provided during the initial operational period.

7.8 Documentation

THERMOPAC will provide complete technical documentation for the plant.

Documentation includes:

- engineering drawings
- equipment manuals
- operation and maintenance manuals
- automation system documentation
- commissioning reports

7.9 Warranty

All supplied equipment will be covered by a warranty period of typically:

12 months from commissioning or 18 months from supply, whichever occurs first.

The warranty covers manufacturing defects and equipment performance under normal operating conditions.

Next Section

The next section should be:

8 Client Scope of Supply

This section defines what the **client must provide**, such as:

- land and buildings
- civil foundations
- utilities
- base oil storage tanks (optional depending on offer)
- plant utilities

10. Exclusions from Scope of Supply

Unless specifically stated in this proposal, the following items are **excluded from the scope of supply of THERMOPAC** and shall be provided by the client.

All equipment and services supplied by THERMOPAC are limited to the **Inside Battery Limits (ISBL)** of the skid-mounted process systems. Any work outside these limits shall be considered **Outside Battery Limits (OSBL)** and is excluded from this offer.

10.1 Civil and Structural Works

The following civil and structural works are excluded from the scope of supply:

- Land acquisition and site preparation

- All civil foundations for process equipment and structures
- Construction of plant buildings, warehouse, laboratory, and office buildings
- Structural steel works and platforms not supplied as part of skid modules
- Pipe rack structures and pipe support structures
- Drainage systems, roads, and paving within the plant premises

10.2 Storage Tanks and Tank Farm

The following tank farm equipment and systems are excluded:

- Base oil storage tanks
- Additive storage tanks
- Finished product storage tanks
- Tank farm piping systems
- Tank farm instrumentation and accessories
- Tank heating systems and insulation

Tank sizing guidelines and design recommendations may be provided during engineering if required.

10.3 Filling and Packaging Systems

Unless specifically included in the commercial scope, the following equipment is excluded:

- Small pack filling lines
- Drum filling machines
- Capping machines
- Induction sealing machines
- Labeling machines
- Packaging conveyors
- Carton packing and sealing machines
- Palletizing systems

These systems may be offered separately if required by the client.

10.4 Interconnecting Mechanical Works

The following mechanical works outside the battery limits of supplied skids are excluded:

- Interconnecting process piping between skid-mounted units
- Product transfer piping between tank farm and blending units
- Utility piping networks
- Piping valves, fittings, and accessories outside skid boundaries
- Pipe racks and pipe supports
- Insulation and cladding of external pipelines

All piping, valves, and accessories **outside THERMOPAC battery limits** shall be supplied and installed by the client.

10.5 Thermic Fluid System Installation

The following items related to the heating system are excluded unless otherwise specified:

- Thermic fluid piping network between heater and process equipment
- Chimney for the thermic fluid heater
- Thermic fluid expansion structure
- Thermic fluid filling and flushing operations
- External insulation of thermic fluid pipelines

10.6 Utility Systems

The following utilities shall be provided by the client:

- Electrical power supply to the plant
- Compressed air supply system
- Cooling tower system (if required)
- Water supply system
- Fuel supply system for the thermic fluid heater

Utility consumption data will be provided during the engineering stage.

10.7 Electrical Infrastructure

The following electrical equipment and infrastructure are excluded from the scope of supply:

Power Distribution Panels

- PCC – Power Control Center
- PMCC – Power Motor Control Center
- MLDB – Main Lighting Distribution Board
- ELDB – Emergency Lighting Distribution Board
- APFCR – Automatic Power Factor Correction Panel
- LPDB – Lighting Power Distribution Board
- LDB – Lighting Distribution Board

Electrical Systems

- Electrical substation and transformers
- Main plant electrical distribution system
- Cable trays and cable trenches outside skid limits
- Interconnecting electrical cabling between plant sections
- Earthing and grounding system for the entire plant
- Plant lighting systems

Electrical wiring and instrumentation **within the supplied skid-mounted equipment** will be included.

10.8 Material Handling Equipment

The following items are excluded unless specifically included:

- Forklifts
- Storage racks
- Warehouse material handling systems
- Internal plant transport equipment

10.9 Raw Materials and Consumables

The following items are excluded from the scope of supply:

- Base oils
- Additive packages
- Thermic fluid oil
- Laboratory chemicals and consumables
- Packaging materials and containers

10.10 Statutory Approvals

The client shall be responsible for obtaining all statutory approvals and permits required for plant installation and operation, including:

- Environmental approvals
- Factory licenses
- Fire safety approvals
- Electrical authority approvals
- Local regulatory permits

10.11 Manpower

The supply of operating personnel, laboratory staff, and plant management personnel is excluded.

THERMOPAC will provide **operator training during commissioning** to ensure proper plant operation.

FINAL PRICE FOR 100TPD BLENDING PLANT

150 TPD					
Sr. No.	Description	Unit price (USD)	UOM	Qty.	Price (USD)
1	Automatic Batch Blending Skid ABB - 20T (ABB 2301 & 2302)	102,095	SET	2	204,190
2	Automatic Batch Blending Skid ABB - 16T (ABB 2303)	95,025	SET	1	95,025
3	Automatic Batch Blending Skid ABB - 10T (ABB 2304 & 2305)	95,851	SET	2	191,701
4	VI Mixer with Mechanical shredder (AMX-VI-2307 16 KL)	128,501	SET	1	128,501
5	Holdup tank with heating arrangement (PF-2401, PF-2402, PF-2403 & PF-2404 20 KL)	15,050	SET	4	60,200
6	Holdup tank with heating arrangement (PF-2405 & PF-2406 10 KL)	15,501	SET	2	31,001
7	Additive tank (T-2108, T-2109 & T-21010 25 KL)	17,850	SET	3	53,550
8	SCAD control Input base oil Manifold (M-101)	17,150	SET	1	17,150
9	SCAD control Output base oil Manifold (M-102)	17,150	SET	1	17,150
10	SCAD control Blending to Finish tank Manifold (M-103)	24,501	SET	1	24,501
11	SCAD control Finish tank to Filling Manifold (M-104)	24,501	SET	1	24,501
12	PLC SCADA Control panel for Blending Plant	89,950	SET	1	89,950
13	16 Drum heating oven with control panel	38,500	SET	1	38,500
Utility					
	Thermic fluid heater				
	Thermic fluid heater 1 Million Kcal/Hr	47,950	SET	1	47,950
	Compressor with Air dryer	6,650	SET	1	6,650
12	Transfer pumps with frame, coupling, coupling guard, and Motor				
	Base oil receipt				
	PMP-101 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-102 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-103 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	Base oil supply				
	PMP-201 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-202 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-203 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-204 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-205 (Johnson Make 30 M3 @ 10-500 Cst)	10,313	SET	1	10,313
	PMP-206 (Johnson Make 15 M3 @ 500-10000 Cst) with VFD	7,957	SET	1	7,957
	PMP-207 (Johnson Make 15 M3 @ 500-10000 Cst) with VFD	7,957	SET	1	7,957
	PMP-208 (Johnson Make 15 M3 @ 500-10000 Cst) with VFD	7,957	SET	1	7,957
	PMP-209 Portable (Johnson Make 15 M3 @ 500-10000 Cst) with VFD	9,832	SET	1	9,832
	PMP-211 (3-5 M3) Eccentric screw container pumps JP-700	6,036	SET	1	6,036
	PMP-212 (3-5 M3) Eccentric screw container pumps JP-700	6,036	SET	1	6,036
	PMP-213 (3-5 M3) Eccentric screw container pumps JP-700	6,036	SET	1	6,036
	PMP-214 7.5kw 2 inch Silverson Inline High Shear Mixer	16,830	SET	1	16,830
	Finish product to filling				
	PMP-401 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592
	PMP-402 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592
	PMP-403 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592
	PMP-404 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592

	PMP-405 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592
	PMP-406 (Johnson Make 30 M3 @ 10-500 Cst) with VFD	10,592	SET	1	10,592
Flow meters					
	FM_101 to FM_107	3,350	SET	8	26,800
	Radar Type Level Transmitters				
	VEGA Make	1,799	SET	13	23,387
FILLING & PACKAGING UNITS					
Line 1 & 2 - For Filling 1 Lts to 5 Lts					
1	Automatic Volufil Machine 5000 - 8X as per annexure 1	184,600	SET	2	184,600
	Extension Conveyor @ 2.4 Mtr			2	
2	Automatic Screw Capping Machine 1X as per annexure 2			2	
	Body size change parts, required for each bottle shape/size			8	
	Cap size change parts required for each cap			2	
3	Automatic Online Induction Cap Sealer as per annexure 3			2	
4	Labeling Machine – Two Sided as per Annexure 4			2	
5	BOPP Carton Tapping with Carton Coding Machine as per Annexure 5			2	
Line 3 - For Filling 20 Lts					
6	Volufil Machine 6X			1	
7	Automatic Lid Pressing Machine as per Annexure 7			1	
8	Packing table			2	
Line 4 - For Drum Filling					
9	Drum Filling Machine - Weightily 210 CPSS as per Annexure 8			1	
	Idler roller conveyor 2 mtr infeed side and 2 mtr out-feed side.			1	
	Packing table			2	
Mechanical item					
Supervision and commissioning					5,000
Design & engineering					5,000
Total Ex-Factory					1,490,000
Freight & Insurance					60,000
Total Amount USD					1,550,000

- Documentation
 - Plot plans and general arrangement and PFD's
 - Material and heat balances
 - Equipment data sheets and specification sheets
 - P&IDs, U&IDs and piping specs and layout drawings
 - Package units specifications
 - Instrumentation drawings
 - Electrical single line diagrams, electrical drawings
 - Insulation & painting specifications
 - Operating manuals
 - Maintenance manuals.
- Pre-commissioning and Commissioning Assistance:
 - 2-week onsite assistance for final hook-up of utilities and instruments and control wiring.
 - 2 weeks onsite assistance for start-up
 - 2 weeks onsite assistance for demonstration and training.
- Thermopac exact Scope in Construction, Erection, Pre-commissioning and Commissioning.
 - Process design and technical know-how
 - Mechanical design, engineering and structural basic design.
 - Electrical basic design
 - Instruments and control systems basic design.
 - Procurement and supply of the plants and equipment's
 - Supervision of construction, erection/installation of the plant
 - Start-up / commissioning of the plant
 - Operation / training to the staff.

Customer Scope:

The following supplies and services are excluded from our scope, which will have to be carried out by the customer at their cost:

4	Tank farm tank fabrication and pipe work at site
5	Fire Fighting

- General site preparation and civil foundation
- Housing for control panels.
- Labour for erection of plant, cranes, chain pulley etc.
- Local transport
- Building and shed
- Water treatment plant for the steam boiler (We can quote after receiving the mineral analysis report of the raw water)
- Water for gland cooling,
- Feed oil, fuel.
- First fill of lubricants and greases for the gearboxes / bearings
- Filters.
- Drum Decanting System
- Air compressor with Air dryer
- Tank level transmitters
- Forklift & hand pallet trucks
- A complete set of laboratory equipment and accessories
- Batch coding system
- Appointing following outside agencies and the cost thereof:
 - Fabrication and piping contractor
 - Licensed electrical contractor

Major Emissions & Effluents

- Emission from Thermic Fluid heater chimney.
- Oily wash water from floor cleaning and spillage from working areas.

Note: The blending process as such does not produce waste water.

Machinery Noise:

- Air Compressor will give out 80 dBA

Key Equipments of the Process

Equipment Description

- Blending tanks
- Additive heating tanks
- Steam boiler
- Mechanical Seals
- Switchgear
- Control Panel

Country of Origin

Thermopac India
Thermopac India
Thermopac India
Eagle, India
Schneider/Equivalent
Thermopac India

Codes and Standards

- American Society of Testing Materials (ASTM)
- American Society of Mechanical Engineers (ASME)
- American Petroleum Institute (API)
- Tubular Exchanger Manufacturer's Association (TEMA)
- Bureau of Indian Standards (BIS)
- Local Standards where the plants will be located

Advantages of the Process

- Environment friendly
- Highly efficient with latest technology
- Minimum contamination and rework
- PLC automated, with SCADA system
- Skid mounted – Expandable, reduces set-up cost and time
- Modular construction
- Handles a variety of lube oils

Shipment Details

Thermopac will be shipping the entire project from their factory near Mumbai, India. Thermopac project cost is based on CIF nearest port to customer plant location. Further transportation to customer project site, loading, unloading and placing of the equipment on foundations are to be arranged by the customer at their cost.

On-stream Time:

The plant is designed for throughput considering a total of 52 weeks per year with 5 working days per week and one shift per day of 8 hours. The plant should be taken up for cleaning/greasing/oiling/scheduled maintenance as follows:

<u>Schedule</u>	<u>No. of days</u>	<u>Total days in year</u>
Monthly	-	-
Quarterly	1	4
Unscheduled		4
Total down time days in a year		8 Days